Rock, Paper, Scissors
Marking Scheme

Marking Key (out of 75 marks)

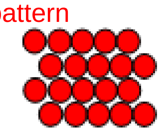
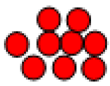
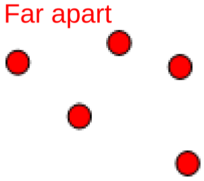
1. Write the name of the state of matter represented by the pictures below.

A B C

____Solid____________ ______Liquid________ _____Gas___________

(1.5 Marks)

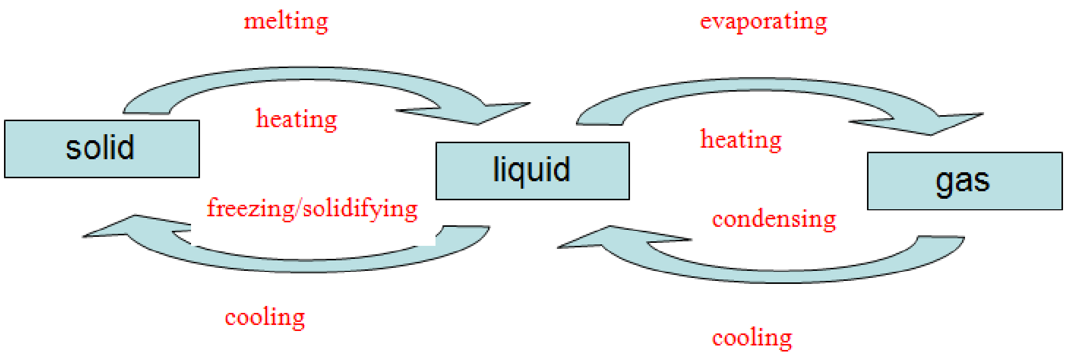
2. Complete the table below.

	Draw the arrangement of particles	**Describe the movement of particles**	**Does it change shape when poured?**	**Does it change volume when the size of the container changes?**	**Can it be compressed?**
solid	Close together and in a regular pattern 	Vibrate on the spot	No	No	No
liquid	Close together and in a irregular pattern 	Move slowly and roll over each other	Yes	No	No
gas	Far apart 	Move fast and collide into each other	Yes	Yes	No
	(3 marks)	**(3 marks)**	**(1.5 marks)**	**(1.5 marks)**	**(1.5 marks)**

3. a) Complete the diagram below by writing for each arrow:

   i) **what causes it to change state**, and

   ii) **the name of the change**.



(8 Marks)

b) Describe what is happening to the particles of matter when changing between each state.

When heated particles start to move faster

As they move faster they collide into each other and move further apart

When cooled particles slow down

As they slow down less collisions occur and they move closer together.

(4 Marks)

4. Changes that occur in substances can be either physical or chemical. What are the differences between physical and chemical changes?

New substance formed (chemical change)

Can't be reversed (chemical change)

(2 Marks)

5. Write beside each letter the type of change (physical or chemical) pictured.

physical change	chemical change
A	B
D	C

(4 marks)

6. If two substances are mixed together, how can you tell if a chemical change has taken place?

Evidence of a new substance formed
1. Bubbles (gas produced)
2. Colour change
3. Precipitation (solid formed)
4. Change in temperature
5. Electricity produced

(5 Marks)

7. Write each of the following substances in the correct column of the table below.

air	water	iron sulfide	zinc
sulfur	a cup of coffee	magnesium	soil
carbon dioxide	salt solution		

Element	Compound	Mixture
sulfur	carbon dioxide	air
magnesium	water	a cup of coffee
zinc	iron sulfide	salt solution
		soil

(10 Marks)

8. Complete the table for each compound listed below.

Formula	Number and name of atoms
HNO_3	1 x hydrogen, 1 x nitrogen, 3 x oxygen
HCl	1 x hydrogen, 1 x chlorine
$CuCO_3$	1 x copper, 1 x carbon, 3 x oxygen
NaOH	1 x sodium, 1 x oxygen, 1 x hydrogen
$CaCl_2$	1 x calcium, 2 x chlorine

(10 Marks)

9. A new anti-rust product called NORUST has just come onto the market. The packaging reads: *Spray NORUST on your nails and dry for 12 hours above 40^{o}C to form a protective anti-rust coat*. Caution: NORUST contains toxic particles.

Choice magazine has received a complaint about the product not working. **(4 marks)**

a) Is NORUST a solid, liquid or gas? Explain your reasoning.

NORUST is a liquid spray. You cannot spray a solid or a gas onto metal nails and expect it to stick. (2 marks)

b) What two safety recommendations would you add to the caution label on the packaging of NORUST?

Being a toxic substance:-

- the user should wear protective gloves
- the user should wear a mask or respirator
- the user should wear protective glasses or goggles
- the user should spray the nails in an open area where there is little danger of breathing in the spray.

(any two of the above reasons.) (2 marks)

10. Design an experiment to test whether the product works – and whether it is likely the customer didn't follow the instructions.

Include a labeled diagram of the set-up as part of the method. **(16 marks)**

Accept any proposal here that addresses the following:-

Method offers a practical way of spraying all sides of the nails. (4 marks)

Method allows drying to occur in an environment above 40°C for 12 hours. (4 marks)

Method incorporates safety aspects noted in 9b. (4 marks)

Labelled diagrams included. (4 marks)